More Than the Risk: Frontier LLM Behavior in AI Development Races Depends on the Rival

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ABSTRACT

Large language models are increasingly used to advise and act on consequential decisions. As such systems become more involved in organisational decision making, an AI-safety question arises: how would frontier agents behave when competition rewards speed while caution reduces catastrophic risk? We study this question in an idealised AI-development race, where agents repeatedly choose between slower safe development and faster unsafe development.

Across the tested frontier routes, risk responses share a qualitative direction but differ in shape. Agents respond to increasing risk, with profiles ranging from gradual adaptation to near-deterministic switching. Behaviour is also strongly opponent-dependent. In a scripted-rival study of three admitted routes, changes in the rival's behaviour move the agents substantially more than changes in the stated risk itself. Thus, self-play outcomes can reflect an interaction between two copies of a policy rather than an intrinsic risk preference.

Frontier agents also show a distinctive relationship to the theoretical and human benchmarks. The four graded routes' responses resemble the near-neutral regime of the evolutionary model more closely than its strong-selection prediction, while their action trajectories occupy a more concentrated behavioural space than those of human participants. These results suggest that AI-mediated competitive decisions may be model-specific, opponent-dependent and behaviourally concentrated, motivating evaluation at the level of interactions and trajectories rather than aggregate action rates alone. Claims are scoped to the tested routes, one prompt version and one idealised game.

KEYWORDS

Large language models, AI development races, multi-agent systems, repeated games, AI safety, opponent-conditioned behaviour

1 INTRODUCTION

LLMs are increasingly studied as agents for decision support and strategic interaction [2, 17]. As organisations consider delegating more strategic tasks to such systems, an AI-safety question follows: if a decision about how fast and how carefully to develop a powerful technology were delegated to a frontier model, what would the model decide, and would its decision look like the ones people make? The question is prospective throughout, and we do not claim that any organisation today hands a safety decision to a language model. We claim only that the behaviour a model would bring to such a decision is measurable now, in a controlled game. Controlled evaluation before widespread deployment can expose interaction effects that would be difficult to isolate in operational settings.

Competition over advanced artificial intelligence can create a safety dilemma: each company may benefit from moving faster, but safety work slows it down when rivals do not follow the same standard [5, 6, 10]. An *AI development race* is a simple game that represents this tension, in which every player chooses each round between slower Safe development and faster Unsafe development [11, 16]. The game is idealised and does not model a real AI company in full. Its value is control: it lets us hold the stated danger, the rival's behaviour and the relative progress fixed one at a time, which no observation of real firms allows. We reproduce the two-player game studied by Domingos and Han [14] and add a compatible multiplayer version based on Han et al. [16]. We call one hosted model endpoint at one fixed configuration a *route*, because the same model name can be served under settings we do not control.

Human experiments show why an average is the wrong thing to compare. In the two-player study of Domingos and Han [14], the preregistered comparison between the two higher private-risk conditions found no clear difference; exploratory analyses instead linked later Unsafe choices to the opponent's previous action, to whether the player was ahead or behind, and to the player's first action, so an early choice changes the later path of the interaction. A useful human and LLM comparison must therefore examine full action sequences, which we call *trajectories*, rather than only the overall Unsafe rate.

Two problems make that comparison harder than it looks. The first is that a correctly formatted action is not an informed one. LLMs already play games of this kind as simulated participants, decision-making agents and game-playing systems [1, 2, 4], yet a model can return a valid action without tracking the state, can predict an opponent's pattern without acting on it [2], and can distort one simulated experiment while reproducing three others [1]; in a repeated game a single such action changes the next state and every later round. Neither a plausible trajectory nor a verbal strategy report is therefore evidence here that the agent tracked the game, and an aggregate UNSAFE rate close to the human one is not evidence that it reproduces human decision dynamics or holds a general safety preference [7].

The second is that prompt-based evaluation is sensitive to presentation choices that do not change what the task asks. Formatting, option order, response-token identity, output-format requests, whitespace and persona all shift model output [24, 26, 27, 29, 33], and strategic choice is no exception: reversing action labels or reassigning their payoffs changes choice frequencies [17], decisions move with topic, actor and world framing [25], and performance falls when a familiar default mapping is replaced by a stated alternative [30]. We therefore test wording, answer order, arithmetic disclosure, narrative framing and opaque response codes separately, and do not treat one familiar presentation as a neutral measuring instrument.

Reproducible game-based frameworks attack the same difficulty from the other side. FAIRGAME standardises repeated-game experiments across models, languages and persona conditions against game-theoretic predictions [8]; later work adds payoff scaling, a Public Goods Game and supervised recognition of canonical strategies, and finds cooperation moving opposite to its evolutionary benchmark [18, 19]. What those designs hold constant is the opponent, because the opponent in them is another copy of the model under test.

One more result shapes what a delegate population should be compared against. Simulated populations recover broad contrasts while showing less heterogeneity than the people they stand in for, in markets, cognitive phenotyping, survey populations and persona-based simulation [9, 12, 13, 15, 28], which is why position papers call social simulation defensible for exploratory use until validated against the population of interest [3]. Strategic-game evidence is similarly unsettled, with policies that shift with parameters and framing, heuristics applied more rigidly than people apply them, cooperation in place of the predicted equilibrium, and action sequences that resist reproduction from prompts alone [21, 22, 31, 32]. A set of delegates that agrees with itself more than people do is a different object from a delegate that is simply more or less cautious, so our human comparison reports within-population diversity and not only aggregate UNSAFE rates.

We therefore ask three questions about the delegate's behaviour. *Risk:* how do frontier agents trade speed against safety as the stated chance of catastrophe rises? *Rivalry:* are those choices a property of the model, or do they emerge from the opponent the model happens to meet? *Humans and theory:* do the agents reproduce the strategic structure that evolutionary game theory predicts, or the one human participants display?

Each answer changes how the next should be read. The agents do respond to the stated danger, within a narrow band, and one route replaces the trade-off by a switch from always fast to always careful. But where we put the agent against scripted code instead of a second copy of itself, the rival's stance moves play several times further than the whole range of stated danger does, and neither external reference shows that shape: the evolutionary model predicts a switch where the agents deliver a gradient, and on raw action sequences human participants use nearly the whole policy space while every admitted route uses a small part of it.

Three contributions follow. First, a scripted-rival design for repeated-game evaluation and the critique it licenses. That an LLM's self-play behaviour need not predict its behaviour against a different agent is known [20, 22], and fixed opponents are standard practice in this literature [23]; what we add is the version that matters for a safety reading, that self-play can hide the response to the treatment the study is about, measured against a rival the agent cannot influence and is never told about. Second, a population-level comparison with human participants read on raw action sequences rather than on aggregate rates, which is what makes the compression of policy visible at all. Third, an auditable race mechanism in which the engine and not generated text resolves every round, entered through a comprehension screen that scores answers to frozen questions against ground truth and never reads a race, so that the set of routes interpreted strategically was fixed by a rule that cannot see the behaviour being interpreted. Exploratory multiplayer results remain in the supplementary material, because the available position analysis covers two checkpoints and cannot identify a general group-size effect.

Every behavioural conclusion is scoped to its tested route, prompt version, decoding contract, provider path and run period. The labels SAFE and UNSAFE denote actions inside the experimental game and are not measurements of general LLM safety, of a subjective risk preference, or of suitability for autonomous deployment, and nothing measured here is a harm, a real safety outcome, or an action taken by an organisation.

2 PRELIMINARIES

We first specify the two-player game reproduced from Domingos and Han [14] (§2.1), then its generalisation to $N > 2$ companies (§2.2), where only the stage payoff changes shape (Figure 1).

2.1 Two-player game

Two companies $i \in \{1, 2\}$ play a repeated race. In every round t, both simultaneously choose an action $a_i^t \in \{S, U\}$ (Safe, Unsafe) and commit to it before either learns the other's choice, and only then does the round resolve (Figure 1a). Resolution has two parts. First, progress accumulates as $P_i^t = P_i^{t-1} + \sigma(a_i^t)$ with $\sigma(S) = 1.0$ and $\sigma(U) = 1.5$, so Unsafe always closes the race faster. Second, the round pays a stage payoff $\pi_i^t = \pi(a_i^t, a_{-i}^t)$, read off the fixed matrix (own action by row, opponent action by column, Figure 1b):

$$\pi = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}. \tag{1}$$

Unsafe strictly dominates Safe at the stage level (row 2 exceeds row 1 in both columns): the only cost of Unsafe development enters later, through accumulated private risk.

The race lasts a minimum of 5 rounds; after every completed round from round 5 onward it terminates with probability 0.2, giving an expected horizon $\mathbb{E}[T] = 9$. Neither company is ever told the horizon, so neither can condition on the current round being the last (Figure 1a). Termination resolves in two further steps. First, the company with the higher cumulative progress P_i^T wins a prize $B = 100$, split 50/50 on an exact tie. Second, and only for that winner or tied winners, an accumulated private setback risk is drawn:

$$q_i(T) = p_r^{\max} \cdot \frac{n_i^U(T)}{T}, q_i(T) \in [0, 1], p_r^{\max} \in \{0.10, 0.60, 0.90\}, \tag{2}$$

where $n_i^U(T)$ counts company i's Unsafe choices over the whole race and $p_r^{\max}$ is a treatment-level constant, assigned once per race, that we vary across three risk levels. If the draw against $q_i(T)$ fails, that company's entire race payoff, accumulated stage payoffs plus the prize, is zeroed. A company that ends behind is never subject to the draw: it keeps its stage payoffs regardless of how much Unsafe it played, but receives no share of the prize (Figure 1a). This asymmetry, in which risk is built by both companies but realised only against whoever is ahead, is what makes Unsafe development a genuine gamble rather than a free speed advantage.

2.2 N-player game

The N-player game changes exactly one thing. Simultaneous sealed actions, the progress increments, the hidden minimum-five-round horizon, the terminal prize $B = 100$, and the risk formula of Equation 2 all carry over unchanged, with the prize split evenly among the companies sharing the highest progress. What cannot carry

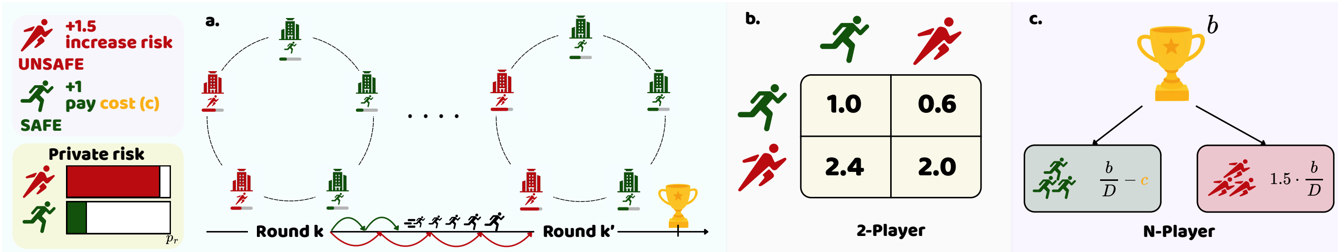


Figure 1: The race and its two payoff views. (a) A repeated development race: each round advances progress by the action-dependent amount, and the winning company receives the prize after the race ends. (b) In the two-player game, the stage-payoff matrix makes UNSAFE attractive within a round, while the private setback risk supplies the safety cost. (c) In the N-player extension, the payoff is divided across the leading companies, so group size changes the payoff structure.

over is the stage-payoff matrix, because with more than two companies there is no single opponent to index it against. Equation 1 therefore generalises to the group-count rule of Han et al. [16], in which both payoffs depend on how many of the N companies chose Safe and not on which ones did (rather than on their identities). The engine supports $N \in \{2, 3, 4, 5\}$. Since the multiplayer runs are an exploratory extension, the rule, its parameters and its results are in the supplementary material.

2.3 Game-theoretic benchmark

The two incentives oppose each other. Stage-level dominance rewards Unsafe immediately, while across the full race Unsafe choices raise the setback risk for a winner or tied winner, so the best action can depend on the risk condition, earlier actions, and the expected response of other players.

We use four simple strategies as a theory benchmark: Always Safe (AS), Always Unsafe (AU), Conditional Safe (CS), which starts Safe and then copies the opponent, and Conditional Unsafe (CAS), which starts Unsafe and then copies the opponent. In the reconstructed finite-population evolutionary model of Domingos and Han [14], the most common strategy changes from AU at $p_r^{\max} = 0.10$, to CAS at 0.60, and to CS at 0.90: the theory predicts not one fixed amount of Unsafe play but a risk-dependent shift between unconditional speed, an aggressive opening and conditional restraint. We use this as a qualitative benchmark, not as a fitted model of LLM output, because prompted self-play trajectories are not samples from an evolutionary population.

3 EXPERIMENTAL DESIGN AND EVIDENCE POLICY

3.1 Agent protocol and units of analysis

At each decision an agent receives a versioned prompt carrying the round, the assigned maximum private risk, the disclosed state, and the preceding revealed action profile when $t > 1$. Same-round decisions are sealed: every response comes from the same pre-action snapshot, before the engine resolves actions, progress, payoffs and risk, so the environment rather than generated text determines all transitions. Raw response, parsed action, retry and parser status, pre-turn state, terminal record, configuration and seed allocation are retained for every decision.

Sampling parameters here are provider-visible contracts, not intended settings. The run manifests record a requested temperature of 0.7 that the hosted SDK forwarded on no baseline route, and no provider confirmed an effective value. Seed handling was route-specific: the SDK stripped the requested seed on the Google routes and forwarded it, application unconfirmed, on the OpenAI and Anthropic ones.

Both seats of a race are the same endpoint, so every gameplay rate below is mirror-match play, until §4.2 fixes the rival instead.

The main behavioural outcome is whether an agent chooses Unsafe. The race, not the individual decision, is the main independent experimental unit, because decisions from one race depend on earlier states and are not separate random samples; uncertainty is grouped by race or matched repetition wherever possible. A *parse failure* is a response that does not follow the required action format, and because the fallback action changes later states, one marks the whole race as contaminated.

3.2 Evidence strata and admission

We separate confirmatory gameplay from diagnostic presentation checks. A *diagnostic pilot* is an exploratory test for possible effects or failures; it does not estimate a stable effect. A *confirmatory* study tests a question under a plan fixed before any result is inspected, with its protocol, route, prompt, configuration, race count and exclusion rules recorded in the run manifest. We never pool the two strata.

A paired behavioural diagnostic compares the canonical prompt against a deterministic four-row decision card that discloses the focal agent's immediate payoff, progress and private-risk consequence for each action profile, while revealing neither the opponent's move nor the stopping horizon. It runs on the one route here whose weights are published, screened by its own battery rather than by the screen of §3.3, so it is read only as a within-route contrast between two presentations.

Three further manipulations are described in full in the supplementary material: a persona framing with a length-matched neutral placebo, eight narrative skins that retell the same game as different stories, and a substitution of the opaque response codes P and Q for SAFE and UNSAFE. The last two are read three ways: a paired first-round comparison at an identical state, a *fixed-state replay* that asks for one action at a previously recorded state without letting the answer change later states, and a *live trajectory* in which each answer does change the next state. A task-validity screen gates which routes enter the behavioural comparison. It is an inclusion rule, not a behavioural outcome, and all gameplay results below are analysed without ranking routes by its scores.

3.3 Quality-control screen on the routes

An action that parses is not evidence that the agent followed the race. Before behavioural analysis, each route therefore answered the same frozen bank of 20 probes over six task-validity domains, three times. A route entered the study only if it cleared the fixed thresholds of 80% overall accuracy, 75% state reconstruction and 75% terminal scoring. Five of nine routes passed; the other four still played the same confirmatory baseline and remain visible in the trajectory-diversity comparison, because two refused routes are the only ones here that reach the human reference range (Figure 5).

The screen reads probes and never reads a race, so gameplay cannot change its verdict. We use that verdict only as a declared inclusion condition, not as a measure of capability or a contrast between admitted and refused routes. The published-weight route used in the arithmetic diagnostic was assessed with a different battery and is not pooled with this screen. Full screening results and robustness checks are retained in the supplementary material.

4 RESULTS

These agents respond to the danger they are told about, and their behaviour is especially sensitive to the rival, takes a shape that the evolutionary benchmark produces near neutrality, and uses a more concentrated set of trajectories than people do. The four subsections take those in turn. A route the quality-control screen of §3.3 refused appears below only where it is named as refused. Three- to five-player races, full model tables, predictive diagnostics and robustness checks are in the supplementary material.

4.1 How agents trade speed against safety

There is a shared directional response across the frontier routes. Every admitted route plays UNSAFE less often as the stated maximum private risk rises, while the shape varies: four of the five give ground gradually and the fifth switches (Figure 2). Each route played 30 races and 558 decisions under one frozen protocol with zero parse failures and ten races in every risk cell, so the difference in shape is not a difference in how much evidence each route supplied.

The four graded routes agree with each other far more closely than their vendors differ. They come from three companies and their risk responses fall in a band 19 points wide, from 38.2 to 57.0 percentage points. We report that band rather than an average over all five, because averaging a switch with four slopes returns 54.8 points, a figure no graded route is near and which one row produces.

That fifth route follows a threshold-like policy. Claude Opus 5 plays UNSAFE on all 186 decisions at risk 0.1 and SAFE on all 372 decisions above it, with no race inside a cell departing from that. Its 100-point figure reaches the largest value on the scale, and because no risk level was run between 0.1 and 0.6, the location of its transition is not measured.

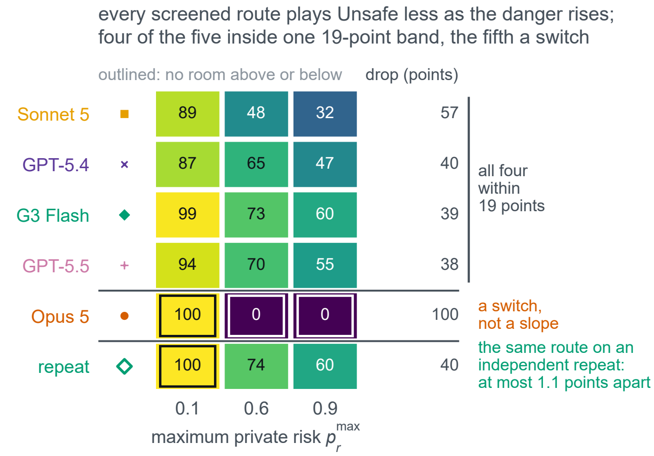


Figure 2: Every admitted route plays UNSAFE less as the stated danger rises. Four of the five graded routes fall within a 19-point band; Claude Opus 5 switches from the ceiling to the floor. The outlined cells are at a boundary and therefore leave no room for a larger or smaller observed rate. The repeat row is the same Gemini 3 Flash route collected under an independent administration.

At the highest risk the game offers, where a winner's whole payoff is zeroed with probability up to 0.9 in proportion to how much UNSAFE it played, the four graded routes still play UNSAFE on roughly a third to three fifths of their decisions against a copy of themselves, in the r=0.9 column of Figure 2. This variation shows that risk-aware behaviour remains contingent on route and interaction context. Two routes can also reach a similar rate through very different sets of races, which §4.4 measures directly.

A route's profile is a property of the route rather than of the run that produced it. Repeating the whole protocol on one route under the same frozen contract moves no risk cell by more than 1.1 points over 186 decisions, against a 93.0-point spread in risk response across the nine routes, and moves its own risk response from 38.7 to 40.3. That bounds run-to-run variation for that route at this sample size and licenses no such bound for a route collected once, which matters here because no provider confirmed applying the requested sampling seed.

Representation-equivalent prompt changes also move rates; diagnostics and the fully crossed symbol-mapping experiment are reported in the supplement.

4.2 The rival can matter more than the stated risk

In the three routes tested against scripted rivals, the observed behaviour does not support a context-independent risk profile. Across these routes, changing who is on the other side moves play much more than changing how dangerous the race is told to be. Every rate reported so far is mirror-match play, in which the rival is the route. One campaign fixes the rival instead.

The rival in that campaign is code, not a model. Three admitted routes each played the four benchmark strategies of §2.3 at all

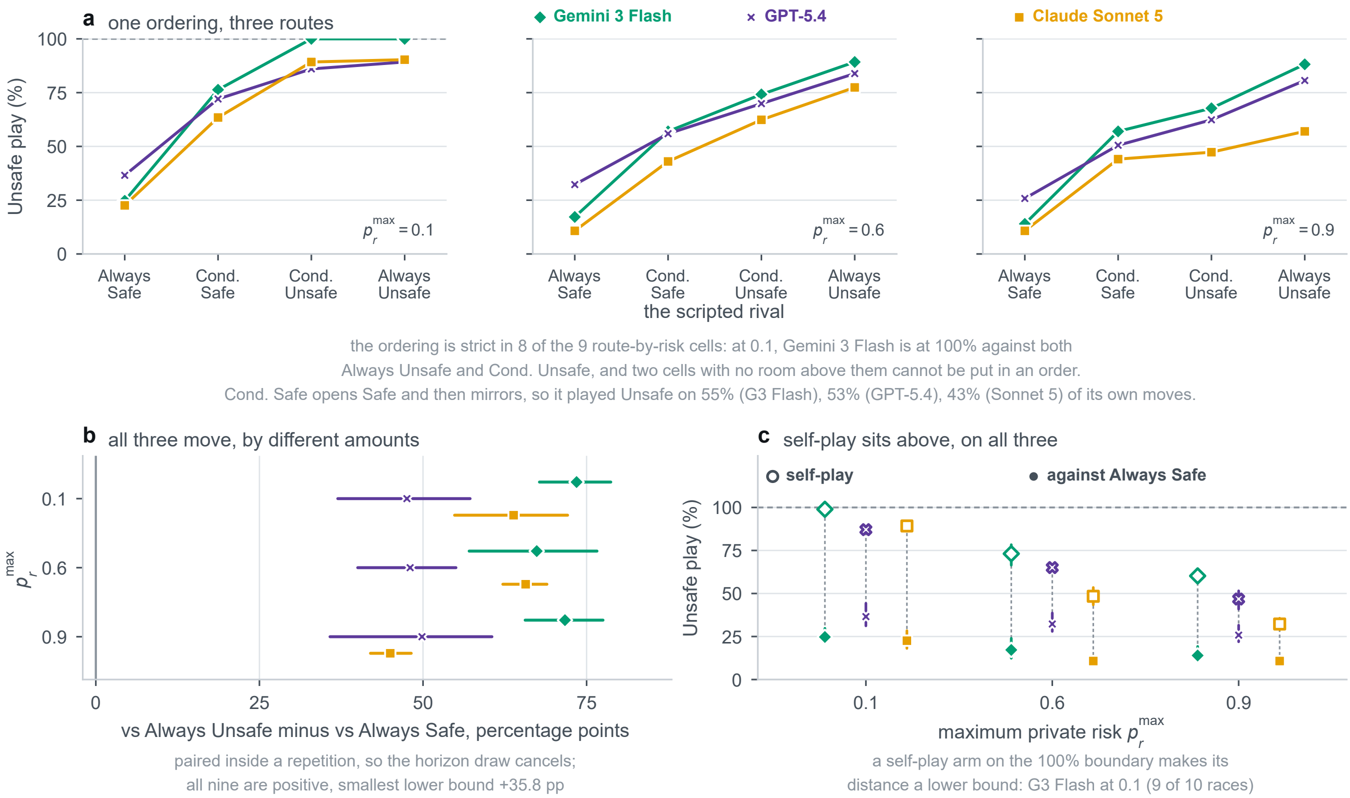


Figure 3: How the rival changes a route's play. The top row shows the four scripted rivals at each risk level, with a rising line meaning that the route responds to the rival's stance. The lower-left panel compares the Always-Unsafe minus Always-Safe contrast within matched repetitions; all nine intervals are positive. The lower-right panel compares each route against a fixed Safe rival and against a copy of itself. Together the panels separate rival-conditioned behaviour from the mirror-match rate.

three risk levels: 36 cells, ten races each, 360 races and 3,348 route decisions with no parse failure anywhere. It is never disclosed to the route, and every one of its 6,696 recorded moves was recomputed afterwards from the route's own recorded history, which no cell failed. The route's seat is counterbalanced five and five inside each cell, and each contrast is taken inside a repetition, so the hidden horizon is differenced out. Because the route cannot influence the rival, the direction of a difference between rivals is not in question.

Who the agent is playing turns out to be worth several times what it is told the danger is, and the two are far enough apart that their intervals do not meet. Both halves are measured the same way and over the same ten blocks, so they can be read against each other directly. Holding the rival fixed at *Always Safe* and moving the stated maximum private risk from 0.1 to 0.9 moves UNSAFE play by 10.4, 12.3 and 12.7 points on Gemini 3 Flash, GPT-5.4 and Claude Sonnet 5. Holding the risk fixed and changing the rival from one that always plays SAFE to one that never does moves it by between 45.0 and 73.5 points. The rival is therefore worth between three and a half and seven times the entire declared range of danger, on every route measured. The comparison survives its own worst case, though not with room to spare. Against the two unconditional rivals, the most the stated risk ever buys is 31.5 points [27.2, 35.3] and the least the rival ever buys is 45.0 [42.0, 48.2], two intervals that do not meet. Adding the two conditional rivals raises the largest risk contrast to 41.0 points [37.7, 43.8], on Claude Sonnet 5 against the rival that opens UNSAFE and then copies, and that one interval does touch the smallest rival contrast. Two risk contrasts are lower bounds because their low-risk arm is at the ceiling in every block, which can only understate the risk side; neither is among the figures quoted here.

Answering the rival is general across these three endpoints. UNSAFE play rises from the rival that always plays SAFE, through the two conditional rivals, to the rival that never plays SAFE, in all nine route-by-risk cells, strictly in eight of them and weakly in the ninth only because two of its cells are at the ceiling (Figure 3a). The paired difference between the two unconditional rivals is positive in every cell, with a smallest lower bound of +35.8 points (Figure 3b). The response to the stated risk survives the change of design as well: against a fixed safe rival, no route's UNSAFE rate rises as the stated risk rises.

How strongly a route answers its rival is a property of the checkpoint. Gemini 3 Flash answers by about seventy points at every risk level, GPT-5.4 by about fifty with an interval that does not meet Gemini 3 Flash's at any risk level, and Claude Sonnet 5 gives a Gemini-sized answer at the two lower risk levels before falling to 45.0 points at the highest. The level differs too, and in the direction that matters for a safety reading: against a rival that never plays UNSAFE, GPT-5.4 still does so 36.6%, 32.3% and 25.8% of the time, more than either of the others at every risk level. These are three separate samples, so intervals that fail to overlap indicate a difference conservatively rather than test one.

A mirror-match rate is therefore an interaction outcome of one policy meeting itself, rather than a stand-alone measurement of how a route treats risk. On all three routes and at all three risk levels the rate against a fixed safe rival sits below the rate the same route reaches against a second copy of itself (Figure 3c), by 74.2, 50.5 and 66.7 points at the lowest risk; on Gemini 3 Flash at that risk the distance is a lower bound, because nine of its ten mirror-match races have every decision UNSAFE. Every mirror-match number in this paper, and every self-play rate in the work it builds on, should be read that way.

The bound on the finding is the design itself. Four reduced strategies are four points in a space of policies rather than a sample of opponents, so this establishes what a route does against *these* rivals, on one game and one prompt version, and three commercial endpoints are not a sample from a population of models.

4.3 The same incentive, a different shape

The agents and the theoretical population move in the same direction under the same incentive, while their response shapes differ. The width of that difference turns on an assumption inside the theory rather than on the game. At the reference setting reported by Domingos and Han [14], selection strength $\beta = 2$ with mutation $\mu = \beta/Z$, the reconstructed evolutionary model predicts a switch: UNSAFE play of 99.2%, 98.0% and 1.9% at maximum risks 0.1, 0.6 and 0.9. The four graded admitted routes decline smoothly, while Claude Opus 5 switches. In the two-route confirmatory block on which this comparison is computed, a separate collection from the nine-route baseline reported above and never pooled with it, Gemini 3 Flash plays UNSAFE 98.9%, 73.1% and 60.2% and Claude Sonnet 5 89.2%, 48.4% and 32.3%, each over 30 races and 558 decisions. The difference is one of shape and not of level: solved on a fine grid of risk values, the benchmark at $\beta = 2$ loses 98 points of UNSAFE play inside a single step of risk 0.005 wide (Figure 4). The supplementary material reports the full five-route fit and inverts the model through the observed rates.

The switch is a property of strong selection, not of the game. We swept selection strength across $\beta \in [0.001, 10]$ crossed with three mutation rules, 90 stationary-distribution cells, each estimated from independent chains with the between-chain spread retained so that a badly mixed cell cannot be mistaken for a result. As β falls towards neutrality the predicted high-risk UNSAFE rate rises off the floor and the model's own risk response becomes graded, which is the shape the routes show. At $\beta = 0.01$ with $\mu = 0.05$ the model predicts 87.3%, 63.9% and 38.0%, the closest cell in the entire sweep to both routes, within 10.3 percentage points of Claude Sonnet 5 and 15.1 of Gemini 3 Flash in root-mean-square error.

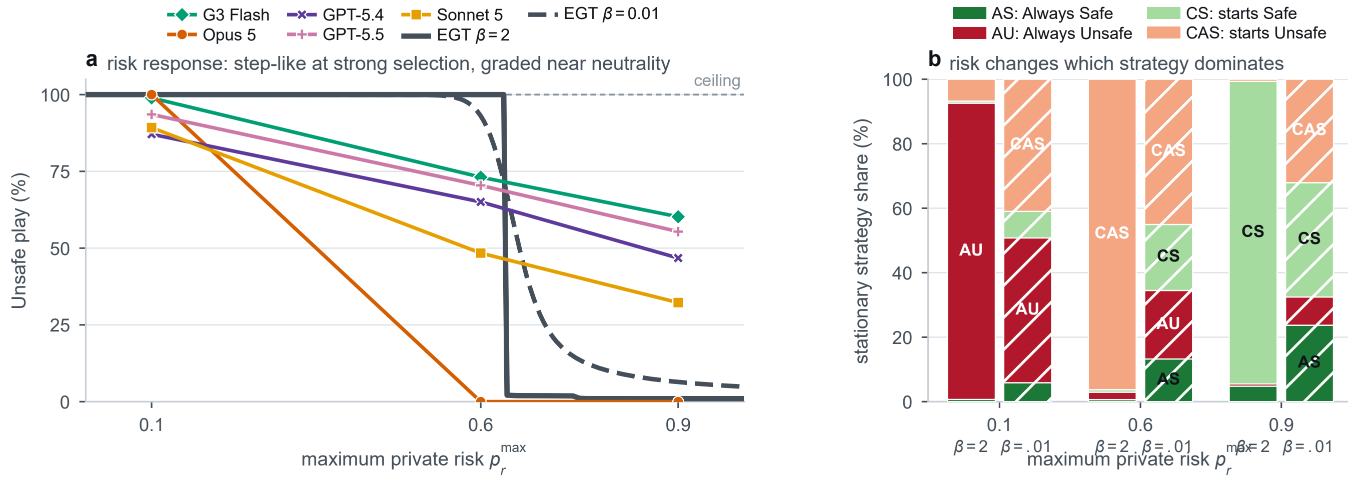
The agents land nearer to the regime that describes the people than to the one usually quoted as the theory's prediction, and that cell is not one we chose to fit: $\beta = 0.01$ and $\mu = 0.05$ are the values the source study reports as its own best fit to human play. An analysis-only extension fits the same existing sweep to all five admitted routes. For the four graded routes, the best well-mixed weak-selection cell has RMSEs of 5.1–15.4 percentage points, compared with 32.9–36.6 points at the reference cell; Claude Opus 5 is the qualitative exception because its observed profile is a near-step switch (100%, 0%, 0%), not a graded decline. The full five-route fit is in the supplement. The interpretation stays bounded. A stationary distribution describes strategy frequencies in a population under selection and mutation, while our measurements are prompted decisions in self-play; the agreement in shape says the graded risk response is consistent with a near-neutral, high-exploration population, not that the routes are evolving. What the sweep establishes is narrower and still useful: a claim that LLM agents contradict the evolutionary benchmark cannot rest on one setting of the selection strength, because the sign of the high-risk gap depends on it.

4.4 Frontier agents are behaviourally concentrated

Twenty people asked to run this race do not behave alike; twenty copies of a frontier route very nearly do. Human participants arrive at their average level of UNSAFE play through many different histories, while each admitted route reuses a small set of them, so a route can match the human mean without resembling the human population at all. We measure this on the action sequences themselves. Each player is one exact paired five-round action sequence, a ten-bit string read within its own risk condition; because the progress increments are fixed, that string is the whole trajectory. The comparison covers all nine routes that faced the screen, admitted and refused alike, on the confirmatory baseline, 60 trajectories per route, against 340 complete human trajectories from the public de-identified dataset of Domingos and Han [14]. That single published study of a single game is the only human reference here, which bounds every human comparison below to that population and that mechanism.

Figure 5 counts how many distinct sequences each population uses out of twenty, against the distribution of 20,000 draws of twenty human participants matched on sample size and risk condition; the human sample is five to seven times larger, so without matching a population could look more diverse merely for being bigger. Twenty-one of the twenty-seven route cells fall below every one of those draws, and the fifteen cells of the five admitted routes are all among them. Against a stricter null of ten complete dyads, which matches the routes' unit of independence and not only their sample size, eighteen do, a count that moves with the draw between sixteen and twenty over forty redraws of that softer null. Humans reach an effective 18.9, 19.7 and 19.3 sequences out of 20 at risks 0.1, 0.6 and 0.9, at mean pairwise distances of 0.47, 0.50 and 0.50. The supplement adds two rarefied statistics, since either alone can mislead: Hill q=1 counts how many sequences a population effectively produces, the mean pairwise Hamming distance how far apart they are. Under the first, every admitted route sits entirely below the human interval at every risk level.

The concentration is strongest for Claude Opus 5, which produces a single sequence, repeated by all 20 of its trajectories, at every risk level, while Claude Sonnet 5 uses two at risk 0.1. Concentration is also distinct from level: Gemini 3 Flash and GPT-5.5



Panel a shows the small-mutation limit; panel b shows the archived finite-mutation compositions at the reference and reported best-fit settings.

Figure 4: The evolutionary benchmark and the admitted route profiles in a common risk coordinate. (a) The small-mutation limit is step-like at strong selection and graded near neutrality, while four routes move gradually and Claude Opus 5 changes sharply between the two lowest tested levels. (b) The corresponding finite-mutation population composition changes its dominant strategy from Always Unsafe through the conditional strategy that starts Unsafe to the conditional strategy that starts Safe as risk rises. The hatched bars show the reported best-fit regime; the unhatched bars show the reference regime.

finish within five points of each other at the highest risk and use a similar number of distinct sequences, 12 and 13, yet the Gemini sequences sit half again as far apart from one another, 0.41 against 0.27 in mean pairwise distance.

Two checkpoints are the exception, and the screen refuses both of them. GPT-5.4 mini uses 20, 17 and 17 distinct sequences out of 20 at risks 0.1, 0.6 and 0.9, and GPT-5.4 nano 16, 19 and 16; both land inside the human null at every risk level, and no other route reaches it once. On this measure they are not less diverse than people. Both are routes the quality-control screen of §3.3 refused, at 75.0% and 51.7% overall probe accuracy. That is why the screen is reported in the body: without it, these two rows would read as the finding that small models match human strategic diversity.

The same picture holds one level down, on the rates themselves. Raw player-level UNSAFE rates, plotted per risk condition in the supplementary material, spread across nearly the whole available range for humans at every risk level while each checkpoint occupies a narrow band of its own. Similarity between a checkpoint's aggregate UNSAFE rate and the human mean therefore does not imply that it reproduces the human distribution. A two-dimensional embedding of the same space is in the supplementary material; we rest no conclusion on it, because recolouring the same coordinates by outcome separates them almost entirely by UNSAFE rate.

Populations that look alike in rate can still be following different rules. Two exploratory analyses in the supplementary material point that way without settling it. A shallow classifier with folds grouped by race recovers the producing population well above a grouped permutation null, and the variables that associate with an UNSAFE choice, read as association and never as cause, separate the populations further: human play is organised mainly by the opponent's previous action, 56% of the attributed importance, and Claude Sonnet 5 is the one checkpoint that leads with the same variable, at 48%. Neither analysis is evidence of a recoverable latent strategy, the human side is the weakly fitted one (ROC AUC 0.63 against 0.97), and both run on an exploratory seven-checkpoint roster containing neither GPT-5.4 nor GPT-5.5, so neither describes the admitted panel and no claim above rests on them.

5 DISCUSSION AND CONCLUSION

These agents incorporate both the stated risk and the rival's stance. Every admitted route plays UNSAFE less often as the stated chance of catastrophe rises, while on the three routes where the opponent was replaced by scripted code, the rival's stance changes behaviour several times more than moving the danger across its whole range. A delegate of this kind is therefore context-sensitive: how the other side is behaving can be more informative than the risk level alone.

A mirror-match rate is best reported as an interaction outcome rather than as an attitude to risk, here or in the work this paper builds on. Scripting the rival costs little and separates a policy from an interaction in this design, making the treatment easier to interpret.

Prompted personas can shift behaviour strongly, reinforcing that instructed context should not be interpreted as a stable latent preference.

The human experiment is an external behavioural reference, not a target an agent should reproduce. Where a route matches a human association, the two populations behave similarly on that one measured aspect of this task; it does not follow that they share a mechanism, and a mismatch does not invalidate the human result.

Four limitations bound the claims directly. Nine commercial endpoints are not a sample from a population of models, and any two of them differ in far more than the one property a comparison names,

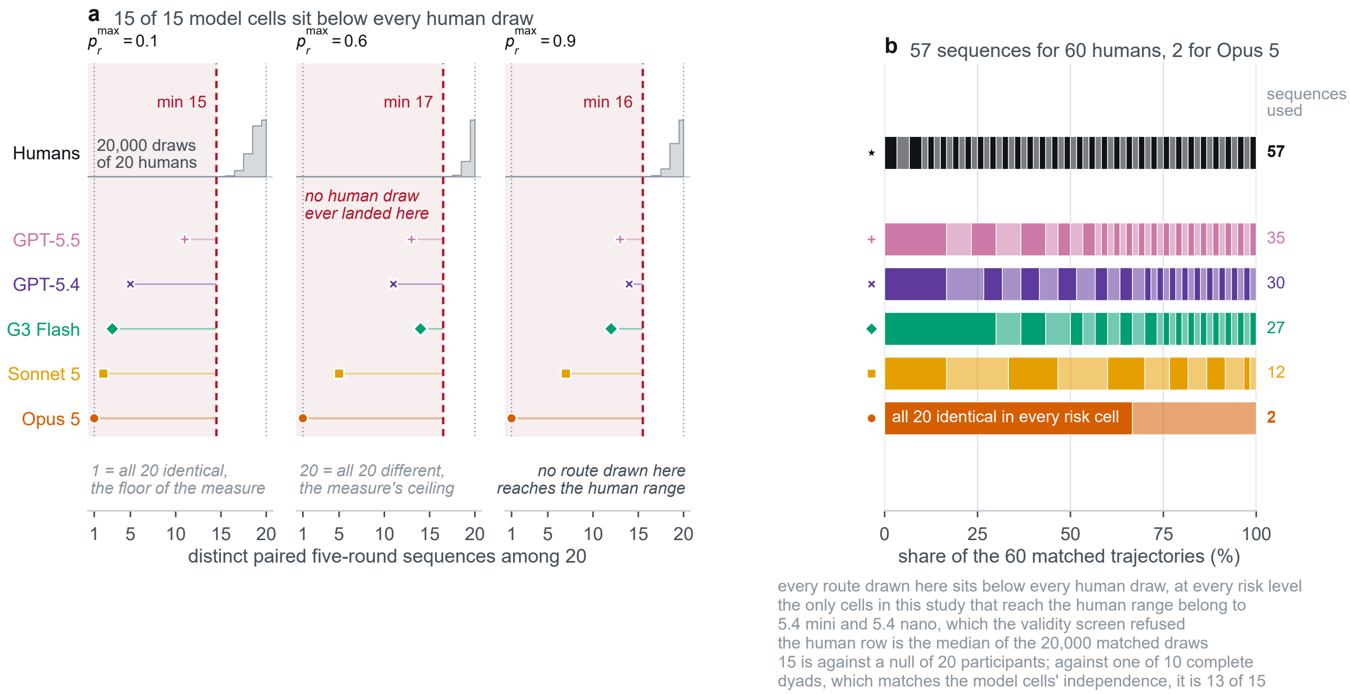


Figure 5: Human participants use far more of the policy space than every checkpoint the screen admits. Panel a places each admitted checkpoint's count of distinct paired five-round sequences against the matched human resampling null; the shaded region is the range no human draw reached. Panel b shows which sequences each population uses.

so every contrast drawn between routes in this paper is descriptive and none of it separates a route's behaviour from its provider, its scale or its training. The scripted-rival campaign covers three routes and four reduced strategies, which are four points in a space of policies rather than a sample of opponents. The human comparison rests on one published study of one game, 340 complete trajectories, and we do not add a second human dataset, because any other dataset would change the game as well as the population. And the evolutionary comparison is a reconstruction audited against pinned EGTtools source rather than a bitwise reproduction of undisclosed author code, so it bounds the regime in which the benchmark is read rather than fitting the agents.

Four further limits are worth stating. Fixed-state replay and live play measure different things, one decision at a saved state against dependent decisions along a trajectory, so their effect sizes are not interchangeable. The representation-robustness design is crossed for two routes, enough to show the effect replicates but not to characterise endpoints in general. A matched group-size grid now exists on one route, but adding a company changes the stage payoff and the prompt length too, so no pure group-size effect is identifiable. Seed forwarding is provider-specific: one route's SDK stripped the requested seed and another forwarded it unconfirmed. And this game models a private speed-versus-safety trade-off and nothing else, with no collective harm, regulation, communication or uncertainty about real capabilities.

The last implication belongs to the population rather than to any single agent, and it stays conditional, because nothing measured here is a deployment. People bring substantial behavioural variety to this race, while copies of one route provide a more consistent profile; which part of the policy space a route settles on differs from route to route. If one model were reused as a strategic adviser across many organisations, its concentrated profile would be one component of the resulting decision environment, alongside the diversity of human decision makers. Whether that balance matters outside this idealised game is not something we can measure here, but it can be measured before delegation is introduced.

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